

CSE 464 - Project Part 3 README (Submission)

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Repository:

[arsule375/CSE464-project-p3] (<https://github.com/arsule375/CSE464-project-p3>)

Primary submission branch: `main`

Feature development branches: `refactor`, `refactor-updated`

Date: May 5, 2026

Build Instructions

Prerequisites:

- Java JDK 11+
- Maven 3.6+

Build command:

```
```bash
cd CSE464-GraphManager
mvn package
```
```

Expected artifact:

```
```text
target/CSE464-GraphManager-1.0-SNAPSHOT.jar
```
```

Latest verified build result:

```
```text
[INFO] Tests run: 35, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
```
```

Feature Summary

Feature implemented for this part: graph search with BFS, DFS, and Random Walk.

| Algorithm | Implementation | Behavior |
|-------------|-------------------------------|--|
| BFS | <code>GraphManager</code> | Breadth-first exploration (level-order) |
| DFS | <code>GraphManager</code> | Depth-first exploration (go deep, then backtrack) |
| Random Walk | <code>RandomWalkSearch</code> | Random unvisited-neighbor walk until success/failure |

Patterns used:

- Template Method: `GraphSearchTemplate`
- Strategy: `SearchStrategy` with algorithm-specific implementations

How To Run BFS / DFS / Random Walk

Option A (Recommended): Demo runner output for all algorithms

```
```bash
cd CSE464-GraphManager
mvn package -q
java -cp "target/CSE464-GraphManager-1.0-SNAPSHOT.jar:${find ~/.m2 -name '*.jar' | tr '\n' ':'}" asu.cse464.DemoTraversalRunner
```
```

Note for Windows: use `;` instead of `:` in classpath separators.

Option B: Run tests

```
```bash
cd CSE464-GraphManager
mvn test
```
```

Test classes:

- `src/test/java/asu/cse464/GraphManagerTest.java` (BFS/DFS)
- `src/test/java/asu/cse464/RandomWalkSearchTest.java` (Random Walk)

Expected Console Output

Input graph (``input.dot``):

```
```dot
digraph G {
 a -> b;
 b -> c;
 c -> d;
 d -> a;
 a -> e;
 e -> f;
 e -> g;
 f -> h;
 g -> h;
}
```
```

Graph shape:

```
```text
a -> b -> c -> d -> (back to a)
a -> e -> f -> h
 e -> g -> h
```
```

BFS output (src=a, dst=h)

```
```text
BFS:
Path Progression: a
Path Progression: a-b
Path Progression: a-e
Path Progression: a-b-c
Path Progression: a-e-f
Path Progression: a-e-g
Path Progression: a-b-c-d
Path Progression: a-e-f-h
Found target node: h
```
```

Final Path: a-e-f-h

...

DFS output (src=a, dst=h)

```text

DFS:

Path Progression: a

Path Progression: a-b

Path Progression: a-b-c

Path Progression: a-b-c-d

Path Progression: a-e

Path Progression: a-e-f

Path Progression: a-e-f-h

Found target node: h

Final Path: a-e-f-h

...

### ### Random Walk output (sample run)

```text

Random Walk (no backtracking, unvisited neighbors only):

Run 1:

Visit Node History: a

Visit Node History: a-e

Visit Node History: a-e-f

Visit Node History: a-e-f-h

Found target node: h

...

Random Walk runs completed: 5

Distinct successful paths found: 2

...

Screenshot Placeholders (Replace Before PDF Export)

```
[INFO] -----
[INFO]  T E S T S
[INFO] -----
[INFO] Running asu.cse464.RandomWalkSearchTest
visiting Path{nodes=[Node{a}]}
visiting Path{nodes=[Node{a}, Node{b}]}
visiting Path{nodes=[Node{a}]}
visiting Path{nodes=[Node{a}, Node{b}]}
visiting Path{nodes=[Node{a}, Node{b}, Node{c}]}
Path{nodes=[Node{a}, Node{b}, Node{c}]}
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.099 s - in asu.cse464
[INFO] Running asu.cse464.GraphManagerTest
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
[INFO] Tests run: 30, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.967 s - in asu.cse464
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 35, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.2.0:jar (default-jar) @ CSE464-GraphManager ---
[INFO] Building jar: /workspaces/CSE464-project-p3/CSE464-GraphManager/target/CSE464-GraphManager.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.662 s
[INFO] Finished at: 2026-05-05T22:20:41Z
[INFO] -----
```

BFS:

Path Progression: a

Path Progression: a-b

Path Progression: a-e

Path Progression: a-b-c

Path Progression: a-e-f

Path Progression: a-e-g

Path Progression: a-b-c-d

Path Progression: a-e-f-h

Found target node: h

Final Path: a-e-f-h

```
1  DFS:
2  Path Progression: a
3  Path Progression: a-b
4  Path Progression: a-b-c
5  Path Progression: a-b-c-d
6  Path Progression: a-e
7  Path Progression: a-e-f
8  Path Progression: a-e-f-h
9  Found target node: h
10 Final Path: a-e-f-h
11
12
```

Random Walk (no backtracking, unvisited neighbors only):

Run 1:

Visit Node History: a

Visit Node History: a-e

Visit Node History: a-e-f

Visit Node History: a-e-f-h

Found target node: h

Run 2:

Visit Node History: a

Visit Node History: a-e

Visit Node History: a-e-f

Visit Node History: a-e-f-h

Found target node: h

Run 3:

Visit Node History: a

Visit Node History: a-e

Visit Node History: a-e-g

Visit Node History: a-e-g-h

Found target node: h

Run 4:

Visit Node History: a

Visit Node History: a-e

Visit Node History: a-e-g

Visit Node History: a-e-g-h

Found target node: h

Run 5:

Visit Node History: a

Visit Node History: a-e

Visit Node History: a-e-f

Visit Node History: a-e-f-h

Found target node: h

Random Walk runs completed: 5

Distinct successful paths found: 2


```

[INFO] Building CSE464-GraphManager 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE464-GraphManager ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE464-GraphManager ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE464-GraphManager ---
[INFO] skip non existing resourceDirectory /workspaces/CSE464-project-p3/CSE464-GraphManager/src
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE464-GraphManager ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:2.22.2:test (default-test) @ CSE464-GraphManager ---
[INFO]
[INFO] -----
[INFO]   T E S T S
[INFO] -----
[INFO] Running asu.cse464.RandomWalkSearchTest
visiting Path{nodes=[Node{a}]}
visiting Path{nodes=[Node{a}, Node{b}]}
visiting Path{nodes=[Node{a}]}
visiting Path{nodes=[Node{a}, Node{b}]}
visiting Path{nodes=[Node{a}, Node{b}, Node{c}]}
Path{nodes=[Node{a}, Node{b}, Node{c}]}
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.134 s - in asu.cse464.
[INFO] Running asu.cse464.GraphManagerTest
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
[INFO] Tests run: 30, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.846 s - in asu.cse464.
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 35, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.720 s
[INFO] Finished at: 2026-05-05T22:21:17Z
[INFO] -----

```

0 Java: Lightweight Mode

Ln 1, Col 1 Spaces:

[GitHub Links \(Commits, Branches, PR/Merge\)](#)

[Refactor commits](#)

| Description | Commit |
|---|---|
| Extract <code>`buildAdjacencyMap`</code> | [3ca6faf] (https://github.com/arsule375/CSE464-project-p3/commit/3ca6faf) |
| Extract <code>`reconstructPath`</code> | [f23d17d] (https://github.com/arsule375/CSE464-project-p3/commit/f23d17d) |
| Extract <code>`buildDotString`</code> | [fe11976] (https://github.com/arsule375/CSE464-project-p3/commit/fe11976) |
| Extract path edge helper into <code>`Path`</code> | [1de7366] (https://github.com/arsule375/CSE464-project-p3/commit/1de7366) |
| Refactor review fixes | [4769f40] (https://github.com/arsule375/CSE464-project-p3/commit/4769f40) |

Random Walk and traversal commits

| Description | Commit |
|--|---|
| Add <code>`RandomWalkSearch`</code> strategy | [9dale83] (https://github.com/arsule375/CSE464-project-p3/commit/9dale83) |
| Address PR #2 review comments | [f8474b2] (https://github.com/arsule375/CSE464-project-p3/commit/f8474b2) |
| Add Random Walk test case updates | [cee170f] (https://github.com/arsule375/CSE464-project-p3/commit/cee170f) |
| Add <code>`DemoTraversalRunner`</code> | [94ff5b7] (https://github.com/arsule375/CSE464-project-p3/commit/94ff5b7) |
| Path test edits | [5d17821] (https://github.com/arsule375/CSE464-project-p3/commit/5d17821) |
| Latest integration update | [fc41adc] (https://github.com/arsule375/CSE464-project-p3/commit/fc41adc) |

Branch links

- Main submission branch:
[main] (<https://github.com/arsule375/CSE464-project-p3/tree/main>)
- Refactor branch:
[refactor] (<https://github.com/arsule375/CSE464-project-p3/tree/refactor>)
- Updated feature branch:
[refactor-updated] (<https://github.com/arsule375/CSE464-project-p3/tree/refactor-updated>)

PR / merge links

- PR #2 (feature branch merged): [Pull Request #2] (<https://github.com/arsule375/CSE464-project-p3/pull/2>)
- Main branch merge sync commit: [694ddf8] (<https://github.com/arsule375/CSE464-project-p3/commit/694ddf8>)

Project Structure

```
```text
CSE464-GraphManager/
|- pom.xml
|- input.dot
|- src/
| |- main/java/asu/cse464/
| | |- GraphManager.java
| | |- GraphSearchTemplate.java
| | |- SearchStrategy.java
| | |- RandomWalkSearch.java
| | |- DemoTraversalRunner.java
| | |- Node.java
| | |- Path.java
| | |- Main.java
| | |- EvidenceGenerator.java
| | \- ProofGenerator.java
| \- test/java/asu/cse464/
| |- GraphManagerTest.java
| \- RandomWalkSearchTest.java
\-- evidence/
 \- search-paths.txt
```
```